Optimal Mixing of Passive Scalars

A Pseudo-Spectral Numerical Study of the
Lin–Thiffeault–Doering Velocity

Adebanji Adelowo

April 27, 2021

Abstract

This thesis investigates the optimal mixing of passive scalars in incompressible fluid flow using the instantaneous-optimal (greedy) velocity field derived by Lin, Thiffeault & Doering (2011). The central objective is to numerically characterise the decay rate of the H^{-1} mix norm under the LTD optimal stirring strategy and to examine how the mixing timescale depends on the spatial scale of the initial scalar distribution.

The scalar transport equation is solved numerically on the two-dimensional torus $[0, 1]^2$ using a pseudo-spectral discretisation (fast Fourier transforms in space) together with an adaptive Dormand–Prince RK45 scheme for time integration. Four families of initial conditions – parameterised by a support-size parameter a – are studied, with L^p norm conservation used as a resolution diagnostic and stopping criterion.

Numerical results confirm that, under the dynamically chosen LTD velocity, the H^{-1} mix norm decays approximately exponentially, consistent with the robust exponential decay reported numerically by Lin, Thiffeault and Doering for the same steepest-descent strategy. Writing this decay as $\|\theta(t)\|_{H^{-1}} \approx \|\theta_0\|_{H^{-1}} e^{-rt}$ with decay-rate magnitude $r > 0$, a representative case ($a = 0.5$) gives $r \approx 0.44$ (mixing timescale $\tau = 1/r \approx 2.27$). Across eight values of $a \in [0.5, 0.9375]$, the fitted decay-rate magnitude scales approximately as $r \propto a^{-1.78}$, appreciably steeper than the a^{-1} dependence characteristic of the Iyer–Kiselev–Xu enstrophy-constrained lower bound. Comparing two resolutions ($N = 32$: exponent ≈ -1.61; $N = 64$: exponent ≈ -1.78) shows this fitted exponent is resolution-dependent and not yet numerically converged; since the higher-resolution estimate moves farther from, not closer to, the benchmark -1 scaling, these two resolutions alone cannot establish whether the discrepancy reflects a finite-resolution artefact, a genuine feature of the greedy strategy, or both.

These results provide a foundation for future studies at higher spectral resolution and for comparison with non-greedy global optimisation strategies.

Keywords: passive scalar mixing, H^{-1} mix norm, pseudo-spectral methods, optimal stirring, Lin–Thiffeault–Doering velocity, incompressible fluid mechanics

Contents

1 Introduction

This report documents a pseudo-spectral numerical simulation of the *optimal mixing problem* for passive scalars on the two-dimensional torus $\mathbb{T}^2 = [0, 1]^2$. The simulation reproduces and extends Gautam Iyer's earlier MATLAB implementation of the instantaneous-optimal (steepest-descent) stirring strategy derived by Lin, Thiffeault and Doering [1].

1.1 Motivation

Mixing a substance efficiently under a fixed stirring budget is a question that appears across scales: from folding cream into coffee, to homogenising reactants in a chemical reactor, to the turbulent dispersal of pollutants and phytoplankton in the ocean. In all of these settings the underlying physics is the same passive-scalar transport equation studied here, and the same question recurs – given a fixed amount of "stirring energy," what motion mixes fastest? The LTD framework answers this question precisely for an idealised, instantaneous notion of optimality, and is a natural entry point into the broader mathematical theory of mixing, which connects functional analysis (Sobolev norms as mixing diagnostics), the calculus of variations (constrained optimisation of the velocity field), and dynamical systems (chaotic advection, filamentation, and the eventual limits imposed by finite resolution or molecular diffusion).

1.2 Goals

- Reproduce the results of Gautam Iyer's earlier MATLAB implementation of this simulation.
- Demonstrate that the H^{-1} mix norm decays exponentially under the Lin–Thiffeault–Doering (LTD) optimal velocity.
- Characterise how the mixing timescale depends on the support size parameter a.
- Derive the LTD velocity rigorously, rather than simply quoting it, so that the numerical implementation can be checked term-by-term against the theory (Section 2).
- Compare the fitted decay-rate scaling in a against the benchmark a^{-1} dependence of the Iyer–Kiselev–Xu enstrophy-constrained lower bound, and test whether the fitted exponent is resolution-converged via a direct two-resolution comparison (Section 6).

1.3 Structure of this Report

Section 2 develops the mathematical framework, including a full derivation of the LTD optimal velocity via constrained optimisation. Section 3 documents the pseudo-spectral numerical scheme. Section 4 describes the initial data families. Section 5 presents the main simulation results at the published resolution $N = 64$. Section 6 adds a resolution-sensitivity study not present in Gautam Iyer's earlier MATLAB implementation. Section 7 closes with conclusions.

2 Mathematical Framework

2.1 The Passive Scalar Equation

We study the advection of a passive scalar $\theta : \mathbb{T}^2 \times [0, \infty) \to \mathbb{R}$ governed by

$$\partial_t \theta + (u \cdot \nabla)\theta = 0, \qquad \theta(\cdot, 0) = \theta_0, \tag{1}$$

where $u : \mathbb{T}^2 \to \mathbb{R}^2$ is a prescribed divergence-free velocity field, $\nabla \cdot u = 0$. The scalar θ is *passive*: it is transported by u but exerts no back-reaction on the flow, and there is no molecular diffusion. Equation (1) states that θ is constant along particle trajectories $\dot{X}(t) = u(X(t), t)$; since the flow map generated by a divergence-free u is measure-preserving, every level set $\{\theta = c\}$ is merely rearranged in time, never created or destroyed. This is the single physical fact underlying everything that follows: mixing is a purely kinematic phenomenon of rearrangement, and the question of "optimal mixing" is really a question about the most effective way to rearrange.

2.2 The Mix Norm

The quality of mixing is quantified by the H^{-1} *mix norm*:

$$\|\theta\|_{H^{-1}}^2 = \sum_{k \neq 0} \frac{|\hat{\theta}(k)|^2}{4\pi^2|k|^2}, \tag{2}$$

where $\hat{\theta}(k)$ denotes the k-th Fourier coefficient of θ. Equivalently, on the mean-zero subspace, $\|\theta\|_{H^{-1}}^2 = -\int_{\mathbb{T}}^2 \theta\, \Delta^{-1}\theta\, dx\, dy$: writing $\phi = \Delta^{-1}\theta$ (i.e. $\Delta\phi = \theta$), integration by parts gives $-\int \theta\phi\, dx\, dy = -\int (\Delta\phi)\phi\, dx\, dy = \int |\nabla\phi|^2\, dx\, dy \geq 0$, confirming that (2) really is a squared norm.

Contrast this with the ordinary L^2 norm, $\|\theta\|_{L^2}^2 = \sum_k |\hat{\theta}(k)|^2$, which weights every Fourier mode equally and is *exactly conserved* by (1) (a direct consequence of u being divergence-free: $\frac{d}{dt}\int \theta^2\, dx\, dy = -2\int \theta\,(u \cdot \nabla\theta)\, dx\, dy = -\int u \cdot \nabla(\theta^2)\, dx\, dy = \int (\nabla \cdot u)\theta^2\, dx\, dy = 0$). The mix norm instead weights the k-th mode by $1/|k|^2$, penalising low wavenumbers (large-scale structure) and suppressing high wavenumbers (fine-scale structure). A scalar field whose L^2 energy sits at low k (a smooth blob) therefore has *large* $\|\theta\|_{H^{-1}}$, while the same L^2 energy redistributed to high k (fine filaments, the same substance stretched thin) has *small* $\|\theta\|_{H^{-1}}$ – even though $\|\theta\|_{L^2}$ is unchanged. This is the precise mathematical sense in which the mix norm captures "mixedness": a rearrangement that conserves L^2 mass while pushing energy to finer scales is exactly what (2) is designed to detect.

2.3 Enstrophy and the Stirring Budget

The "stirring budget" available to the velocity field is expressed as a constraint on its *enstrophy*,

$$\mathcal{E}(u) := \|\nabla u\|_{L^2}^2 = \int_{\mathbb{T}}^2 |\nabla u_x|^2 + |\nabla u_y|^2\, dx\, dy. \tag{3}$$

For a divergence-free planar field this coincides (up to the definition's normalisation) with the L^2 norm of the scalar vorticity $\omega = \partial_x u_y - \partial_y u_x$: writing u in terms of a streamfunction ψ via $u = \nabla^\perp\psi = (-\partial_y\psi, \partial_x\psi)$ gives $\omega = \Delta\psi$ and $\|\nabla u\|_{L^2}^2 = \|\omega\|_{L^2}^2$ after an integration by parts. Bounding $\|\nabla u\|_{L^2}$ therefore bounds the total squared vorticity of the flow – physically, a cap on how vigorously and how finely the flow is allowed to swirl. This is a stronger constraint than capping the kinetic energy $\|u\|_{L^2}$ alone: for fixed enstrophy, a velocity field concentrated at high wavenumber (fine swirls) uses up its budget faster than one concentrated at low wavenumber, since $\|\nabla u\|_{L^2}^2 = \sum_k 4\pi^2|k|^2|\hat{u}(k)|^2$.

2.4 The Lin–Thiffeault–Doering Optimal Velocity

Among all incompressible velocity fields satisfying the enstrophy constraint $\|\nabla u\|_{L^2} \leq F$, the one that minimises $\frac{d}{dt}\|\theta\|_{H^{-1}}^2$ at each instant is the *Lin–Thiffeault–Doering (LTD) optimal velocity* [1]:

$$v = -\Delta^{-1}P\big(\theta\, \nabla\Delta^{-1}\theta\big), \qquad u = F\, \frac{v}{\|\nabla v\|_{L^2}}. \tag{4}$$

Here P is the *Leray projection* onto divergence-free vector fields:

$$Pg = g - \nabla\Delta^{-1}(\nabla \cdot g). \tag{5}$$

The Leray projection removes the curl-free (gradient) component of a vector field g, leaving only its divergence-free component; this is precisely what is required, since any admissible u must itself be divergence-free. The final rescaling by $\|\nabla v\|_{L^2}^{-1}$ enforces the enstrophy constraint exactly (with equality) at each instant – intuitively, there is never a reason to use less than the full stirring budget, since using more of a fixed direction can only decrease $\|\theta\|_{H^{-1}}^2$ faster.

2.5 Derivation of the LTD Velocity

Formula (4) is not an ansatz; it is the exact solution of a constrained optimisation problem, and deriving it makes transparent both why the five-step algorithm of Section 3.2 has the structure it does, and when it can fail (Section 2.6).

Write $\phi := \Delta^{-1}\theta$ (so $\Delta\phi = \theta$), and recall from Section 2.2 that $\|\theta\|_{H^{-1}}^2 = -\int \theta\phi\, dx\, dy$. Differentiating in time and using the symmetry of Δ^{-1} (so that $\int \theta\dot{\phi}\, dx\, dy = \int \dot{\theta}\phi\, dx\, dy$),

$$\frac{d}{dt}\|\theta\|_{H^{-1}}^2 = -2\int \dot{\theta}\,\phi\, dx\, dy = 2\int (u\cdot\nabla\theta)\,\phi\, dx\, dy, \tag{6}$$

using (1). Since u is divergence-free, $\nabla\cdot(u\theta\phi) = (u\cdot\nabla\theta)\phi + \theta(u\cdot\nabla\phi)$, and the divergence integrates to zero on the torus, so $\int (u\cdot\nabla\theta)\phi\, dx\, dy = -\int \theta(u\cdot\nabla\phi)\, dx\, dy$. Hence

$$\frac{d}{dt}\|\theta\|_{H^{-1}}^2 = -2\int u\cdot\underbrace{(\theta\nabla\phi)}_{=:g}\, dx\, dy = -2\int u\cdot g\, dx\, dy. \tag{7}$$

This is exactly the quantity $g = \theta\nabla\Delta^{-1}\theta$ computed in Step 1 of Section 3.2. Because u is divergence-free, the curl-free part of g contributes nothing to the pairing ($\int u\cdot\nabla f\, dx\, dy = -\int(\nabla\cdot u)f\, dx\, dy = 0$ for any scalar f), so $\int u\cdot g\, dx\, dy = \int u\cdot Pg\, dx\, dy$, and (7) becomes

$$\frac{d}{dt}\|\theta\|_{H^{-1}}^2 = -2\int u\cdot Pg\, dx\, dy. \tag{8}$$

Now introduce $v := -\Delta^{-1}Pg$ as in (4), so that $Pg = -\Delta v$, and

$$\int u\cdot Pg\, dx\, dy = -\int u\cdot\Delta v\, dx\, dy = \int \nabla u : \nabla v\, dx\, dy, \tag{9}$$

again by integrating by parts (component-wise, using that both u and v are periodic and divergence-free). Substituting into (8),

$$\frac{d}{dt}\|\theta\|_{H^{-1}}^2 = -2\int \nabla u : \nabla v\, dx\, dy. \tag{10}$$

Proposition 1 (Optimality of the LTD velocity)**.** *Among all divergence-free u with $\|\nabla u\|_{L^2} \le F$, the quantity (10) is minimised by $u = F\,v/\|\nabla v\|_{L^2}$.*

Proof. By the Cauchy–Schwarz inequality applied to the L^2 inner product of gradients, $\int \nabla u : \nabla v\, dx\, dy \le \|\nabla u\|_{L^2}\|\nabla v\|_{L^2} \le F\|\nabla v\|_{L^2}$, with equality iff ∇u is a non-negative scalar multiple of ∇v with $\|\nabla u\|_{L^2} = F$, i.e. $u = F\,v/\|\nabla v\|_{L^2}$ (up to an additive constant, which vanishes on the mean-zero torus). This maximises $\int \nabla u : \nabla v\, dx\, dy$, hence minimises $-2\int \nabla u : \nabla v\, dx\, dy = \frac{d}{dt}\|\theta\|_{H^{-1}}^2$. □

At this optimum, (10) evaluates to $\frac{d}{dt}\|\theta\|_{H^{-1}}^2 = -2F\|\nabla v\|_{L^2} \le 0$, confirming the monotonic decrease of the mix norm asserted informally in Section 2.2, *provided* $\|\nabla v\|_{L^2} > 0$.

2.6 Degeneracy: the First-Order-Critical Case

The derivation above shows that the LTD velocity is well defined, and strictly decreases the mix norm, exactly as long as $v \neq 0$, i.e. as long as $Pg \neq 0$. If θ reaches a state where $g = \theta\nabla\Delta^{-1}\theta$ is already curl-free (so that $Pg = 0$), then *no* divergence-free velocity, however large its enstrophy, can decrease $\|\theta\|_{H^{-1}}^2$ at that instant [1]: such a θ is a critical point of the mix-norm functional under incompressible rearrangement. Lin, Thiffeault and Doering address this degenerate case by instead seeking the incompressible velocity that maximises the growth rate of the *second* time derivative of the mix norm, which reduces to an eigenvalue problem for the minimum eigenvalue of a linear operator built from θ and $\varphi = \Delta^{-1}\theta$ [1]. The numerical implementation here does not solve that secondary eigenvalue problem; it instead monitors for proximity to the degeneracy by comparing $\|\nabla v\|_{L^2}$ against $\|\Delta^{-1}g\|_{L^2}$ and emitting a diagnostic warning if the former is much smaller than the latter relative to a tolerance. In practice this warning was not observed to trigger for the smooth, well-supported initial data families used in Section 4.

Remark 1. The LTD velocity is the *greedy* optimal: it minimises the instantaneous rate of change of the mix norm, not a global functional over time (Proposition 1 is a pointwise-in-time statement). This instantaneous optimality does not by itself establish global optimality over a finite or infinite time horizon: Lin, Thiffeault and Doering themselves note that their local-in-time strategy leaves room for further improvement, and identify whether a global-in-time (optimal control) strategy could close the gap between their rigorous bounds and their simulations as a major open question [1]. Section 5.5 below reports a numerical gap between the fitted decay-rate magnitude observed under the greedy LTD velocity and the support-size scaling suggested by the Iyer–Kiselev–Xu enstrophy-constrained lower bound [2]; Section 6 shows that this gap is entangled with finite-resolution effects and should not, on its own, be read as proof of greedy sub-optimality.

3 Numerical Methods

3.1 Spectral Discretisation

We discretise $[0,1]^2$ on an $N \times N$ grid with spacing $dx = 1/N$, using the DFT convention

$$\hat{f}[k_y, k_x] = \sum_{i=0}^{N-1}\sum_{j=0}^{N-1} f[i,j]\, e^{-2\pi i(ik_y + jk_x)/N}. \tag{11}$$

Row index i corresponds to the y-direction and column index j to the x-direction. The centred wavenumbers are $k_{\text{eff}}[j] = j - N \cdot \mathbf{1}[j > N/2]$, giving the range $\{0, 1, \dots, N/2 - 1, -N/2 + 1, \dots, -1\}$.

Derivative multipliers. Differentiation in physical space corresponds to multiplication in Fourier space by

$$D_x(k_x) = 2\pi i\, k_d(k_x), \qquad D_y(k_y) = 2\pi i\, k_d(k_y), \tag{12}$$

where k_d is identical to k_{eff} except that the Nyquist mode $k = N/2$ is set to zero (to avoid aliasing in first derivatives; see Section 3.3).

Inverse Laplacian. The discrete Fourier realisation of Δ^{-1} is the multiplier

$$\Delta^{-1}(k_x, k_y) = \begin{cases} \dfrac{-1}{(2\pi)^2(k_x^2 + k_y^2)} & (k_x, k_y) \neq (0, 0), \\ 0 & (k_x, k_y) = (0, 0). \end{cases} \tag{13}$$

The Nyquist mode is *not* zeroed here, since the Laplacian is a second-order (even) operator for which the sign ambiguity of the Nyquist frequency does not arise.

Mix-norm weight.

$$\Lambda^{-1}(k_x, k_y) = \frac{1}{2\pi|k|} = \sqrt{-\Delta^{-1}(k_x, k_y)}. \tag{14}$$

Parseval identity. For the unitary DFT normalisation used here, $\left\|\hat{f}\right\|_{\ell^2} = N\,\|f\|_{\ell^2}$, so

$$\|f\|_{L^2} = \frac{\left\|\hat{f}\right\|_{\ell^2}}{N^2}. \tag{15}$$

This conversion factor is used throughout whenever a quantity computed in Fourier space (via ℓ^2 norms of coefficient arrays) must be compared against, or combined with, a continuum L^2 quantity.

3.2 Pseudo-Spectral Evaluation of the RHS

At each RK45 stage the following five steps compute $\partial_t\hat{\theta}$. The alternation between Fourier space (for the linear operators ∇ and Δ^{-1}, applied as cheap pointwise multiplications) and physical space (for the pointwise *products* that make up the nonlinearity) is the defining feature of the pseudo-spectral method: representing a product as a convolution directly in Fourier space would cost $O(N^4)$ per evaluation, whereas transforming to physical space, multiplying pointwise, and transforming back costs only $O(N^2 \log N)$ via the FFT.

1. **Compute** $g = \theta\,\nabla\Delta^{-1}\theta$ in physical space (pseudo-spectral):

   $$\hat{g}_x = \mathcal{F}\Big[\theta \cdot \mathcal{F}^{-1}(D_x \cdot \Delta^{-1} \cdot \hat{\theta})\Big], \quad \hat{g}_y = \mathcal{F}\Big[\theta \cdot \mathcal{F}^{-1}(D_y \cdot \Delta^{-1} \cdot \hat{\theta})\Big].$$

   This is exactly the quantity g appearing in the derivation of Section 2.5.

2. **Project** onto divergence-free fields (Leray):

   $$\widehat{Pg}_x = \hat{g}_x - D_x \cdot \Delta^{-1} \cdot (D_x \cdot \hat{g}_x + D_y \cdot \hat{g}_y),$$

   and similarly for the y-component.

3. **Compute** $v = -\Delta^{-1}Pg$:

   $$\hat{v}_x = -\Delta^{-1} \cdot \widehat{Pg}_x, \qquad \hat{v}_y = -\Delta^{-1} \cdot \widehat{Pg}_y.$$

4. **Normalise** to enforce the enstrophy constraint via Parseval:

   $$\|\nabla v\|_{L^2} = \frac{1}{N^2}\sqrt{\|D_x\hat{v}_x\|_{\ell^2}^2 + \|D_x\hat{v}_y\|_{\ell^2}^2 + \|D_y\hat{v}_x\|_{\ell^2}^2 + \|D_y\hat{v}_y\|_{\ell^2}^2}, \quad u = F\,\frac{v}{\|\nabla v\|_{L^2}}.$$

   This is Proposition 1 applied numerically: v is rescaled to have exactly enstrophy F.

5. **Advect**:

$$\frac{d\hat{\theta}}{dt} = -\mathcal{F}\Big[u_x \cdot \mathcal{F}^{-1}(D_x \cdot \hat{\theta}) + u_y \cdot \mathcal{F}^{-1}(D_y \cdot \hat{\theta})\Big] .$$

Each RHS evaluation therefore requires ten $N \times N$ FFTs (three forward and seven inverse transforms, counting the intermediate transforms in Steps 1–5), each costing $O(N^2 \log N)$, for a total cost of $O(N^2 \log N)$ per stage – independent of N only up to the logarithmic factor, which is why doubling N from 32 to 64 increases per-step cost by roughly a factor of 4–5 (four times the grid points, times a modestly larger logarithm), consistent with the timings reported in Section 6.

3.3 Aliasing and the Absence of Explicit Dealiasing

The nonlinearity in Step 1 above is a *pointwise product* of two fields, each itself resolved on N Fourier modes per direction. In exact arithmetic such a product generates Fourier content up to wavenumber $N-1$ (twice the input bandwidth), but the discrete grid can only represent wavenumbers up to $N/2$; the un-representable high-wavenumber content is *aliased* back onto low wavenumbers, contaminating the low modes with spurious energy. The classical remedy in pseudo-spectral codes is the "2/3-rule": pad the field with zeros to $3N/2$ modes before transforming, multiply, and truncate back to the original $N/2$ bandwidth, exactly eliminating the aliased content at the cost of roughly doubling the FFT size.

The implementation does *not* apply the 2/3-rule (nor any other explicit dealiasing filter beyond zeroing the Nyquist mode for first derivatives, which addresses a different, sign-ambiguity issue, not aliasing). Instead, the resolution-check event of Section 3.5 provides an *a posteriori* diagnostic of loss of resolution: because under-resolution – aliasing error among other possible causes – corrupts the conserved L^p norms of θ, its cumulative effect is what the resolution check is designed to detect, and integration is halted the moment that effect becomes non-negligible. This monitor does not *prevent* aliasing error from occurring and is not a substitute for explicit dealiasing; it only bounds how long the simulation is allowed to run once accumulated resolution loss becomes detectable. The consequence – a stopping time that depends on resolution, and hence a fitted mixing-rate exponent that depends on resolution – is examined directly in Section 6 and is relevant to the deviation from the Iyer–Kiselev–Xu benchmark a^{-1} scaling discussed in Section 5.5.

3.4 Time Integration

The ODE is integrated with an explicit, adaptive-step embedded Runge–Kutta pair of Dormand–Prince type (order 5(4)), with relative and absolute error tolerances of 10^{-6} and 10^{-8} respectively: the solution is advanced with a 5th-order formula while a simultaneously computed 4th-order formula provides a local error estimate, and the step size is adjusted so that this estimate stays within tolerance. No dealiasing-related CFL restriction is imposed explicitly; the adaptive step controller implicitly shrinks the step size as θ develops sharper gradients under stirring, which is the mechanism by which the integrator remains stable as filamentation proceeds, right up until the resolution check intervenes.

Because the solver requires a real-valued state vector, the complex Fourier array $\hat{\theta} \in \mathbb{C}^{N^2}$ is stored as a real vector $y = [\mathrm{Re}(\hat{\theta}_{\text{flat}}), \mathrm{Im}(\hat{\theta}_{\text{flat}})] \in \mathbb{R}^{2N^2}$.

3.5 Resolution Check and Stopping Criterion

Under incompressible advection all L^p norms of θ are conserved. We monitor the event function

$$e(t) = \max\left(| \, \|\theta\|_{L^2} - 1|, \ \left| \frac{\|\theta\|_{L^4}}{\|\theta_0\|_{L^4}} - 1 \right|, \ \left| \frac{\|\theta\|_{L^8}}{\|\theta_0\|_{L^8}} - 1 \right| \right) - \varepsilon, \quad \varepsilon = 10^{-3}. \tag{16}$$

Integration stops when $e(t)$ crosses zero upward, indicating that the grid can no longer resolve the fine-scale features of θ. Using three different exponents ($p = 2, 4, 8$) rather than a single one is deliberate: the higher-order norms are more sensitive to sharp, localised features (large $|\theta|$ concentrated on a small area) than L^2, so drift is typically detected first in L^8, giving an early warning before the coarser L^2 budget is visibly affected – consistent with the observation in Section 5.3 that all three norms begin drifting at essentially the same instant, since by that point the under-resolved filaments are thin enough to affect every norm simultaneously.

4 Initial Data

Four families of initial data are used, all L^2-normalised ($\|\theta_0\|_{L^2} = 1$) and parametrised by a scale parameter $a \in (0, 1)$ that controls the support size. After construction the support is circularly shifted to the centre of $[0, 1]^2$ so that the mixing dynamics are not biased by proximity to the periodic boundary.

Family	Support	Description
I: Sine bump	$[0, a]^2$	$\sin(2\pi x/a)\sin(2\pi y/a)$
II: Diagonal pair	Two sub-squares of $[0, a]^2$	Antisymmetric pair, broken symmetry
III: Strip	$[0, a] \times [0, a/2]$	Aspect-ratio 2:1 strip
IV: Trigonometric polynomial	$[0, 1]^2$	$\sum_{k=1}^{2} c_k \sin(2k\pi x/a)\sin(2k\pi y/a)$

All results in Section 5 use the sine-bump family (Family I), the simplest of the four, so that the effect of a on the mixing rate is not confounded by the initial data's internal geometry; the other three families are used for testing sensitivity to the initial condition's shape (a direction identified as future work in Section 7).

Figure 1 shows all four functions for $a = 0.5$.

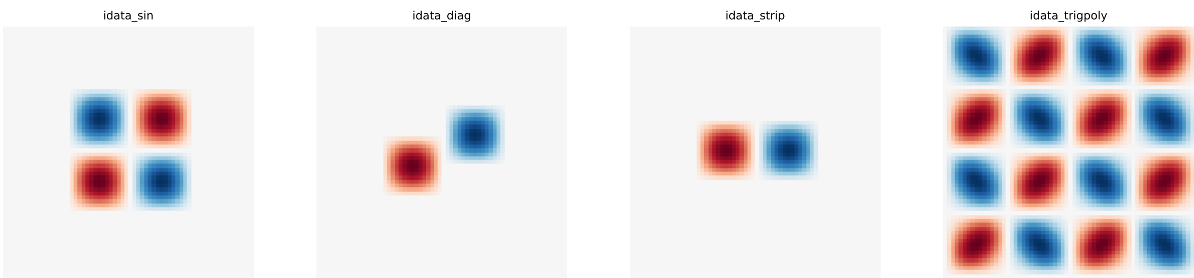

Figure 1: The four families of initial data θ_0 with scale parameter $a = 0.5$ on the $N = 64$ grid. Colour encodes the scalar value (red > 0, blue < 0, white $= 0$).

5 Simulation Results

All results in this section use $N = 64$ spectral modes per direction and enstrophy constraint $F = 1$. Each simulation starts at $t = 0$ and runs until the resolution check event fires. Every figure and number in this section was produced by the accompanying code and independently re-verified: a clean rerun of the pipeline reproduces the values originally recorded in Table 1 to four decimal places in every entry, confirming that the pipeline is deterministic and reproducible end to end for the tested software environment.

5.1 Spatial Evolution of the Scalar Field

Figure 2 shows six snapshots of the scalar field $\theta(x, y, t)$ for $a = 0.5$, sampled evenly between $t = 0$ and the stopping time $t = 1.30$. The LTD velocity progressively stirs θ into finer and finer filaments, consistent with exponential decay of the mix norm. Qualitatively, the sequence illustrates the classical picture of chaotic advection: smooth initial structure is repeatedly stretched and folded, and the length of the scalar's interface grows while its cross-sectional width shrinks, which is precisely the geometric process that pushes L^2 energy to higher wavenumbers and hence drives $\|\theta\|_{H^{-1}}$ down.

Figure 2: Snapshots of the passive scalar $\theta(x, y, t)$ at six equally spaced times for $a = 0.5$, $N = 64$, $F = 1$. The LTD velocity continuously stretches and folds the scalar into progressively finer structures.

5.2 Mix Norm Decay

Figure 3 shows $\log(\|\theta(t)\|_{H^{-1}} / \|\theta_0\|_{H^{-1}})$ for $a = 0.5$. After a brief initial transient the decay is well approximated by a linear fit (dashed), confirming approximately exponential decay:

$$\|\theta(t)\|_{H^{-1}} \approx \|\theta_0\|_{H^{-1}}\, e^{-rt}, \qquad r > 0. \tag{17}$$

Here r denotes the positive decay-rate magnitude, so that the fitted slope of $\log \|\theta\|_{H^{-1}}$ versus t equals $-r$ and the mixing timescale is $\tau = 1/r$. Fitting the slope over the *entire* trajectory gives slope ≈ -0.394 (decay-rate magnitude $r \approx 0.394$); fitting only the *last two-thirds* (the methodology used for Table 1 and Figure 6, which discards the initial transient before the exponential regime is established) gives the steeper slope ≈ -0.440 ($r \approx 0.440$) quoted in Table 1. The two numbers are not in conflict – they measure the average slope over different portions of a curve that is only *approximately* log-linear – but the sensitivity of the fitted rate to the choice of fitting window is itself informative: it shows that the decay, while broadly exponential, has curvature that a single global rate does not fully capture, most visible during the initial transient before the flow has had time to establish fully developed filamentation.

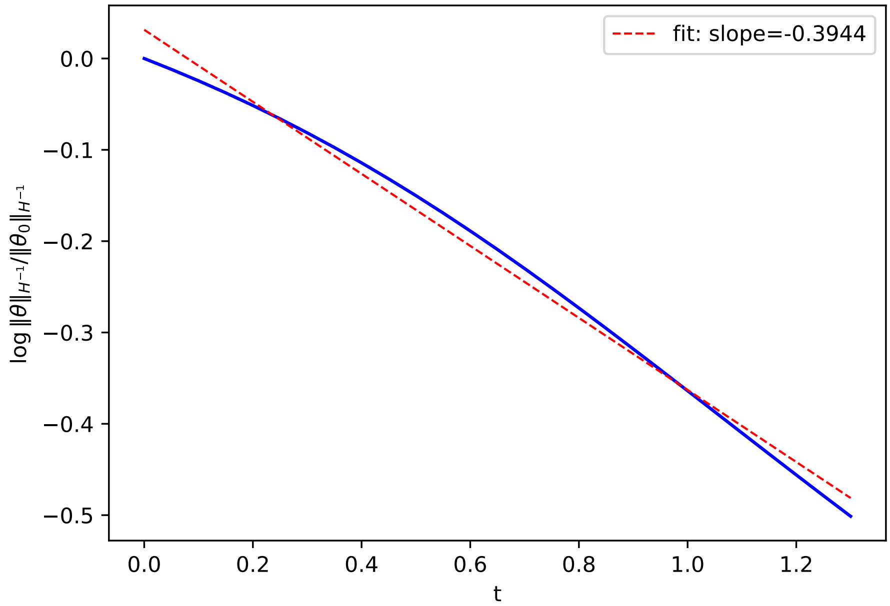

Figure 3: Log of the H^{-1} mix norm vs. time for $a = 0.5$. The dashed line is the least-squares linear fit over the full trajectory, giving slope -0.3944.

5.3 Conservation Law Verification

Figure 4 confirms that the L^2, L^4 and L^8 norms remain ≈ 1 throughout the integration. The norms begin to drift near $t = 1.3$, at which point the resolution check terminates the simulation. The maximum L^2 drift observed is $\max_t |\,\|\theta\|_{L^2} - 1| \approx 3.15 \times 10^{-5}$, two orders of magnitude within the tolerance $\varepsilon = 10^{-3}$ – indicating that the resolution check fires promptly once drift becomes appreciable, rather than allowing substantial error to accumulate silently beforehand.

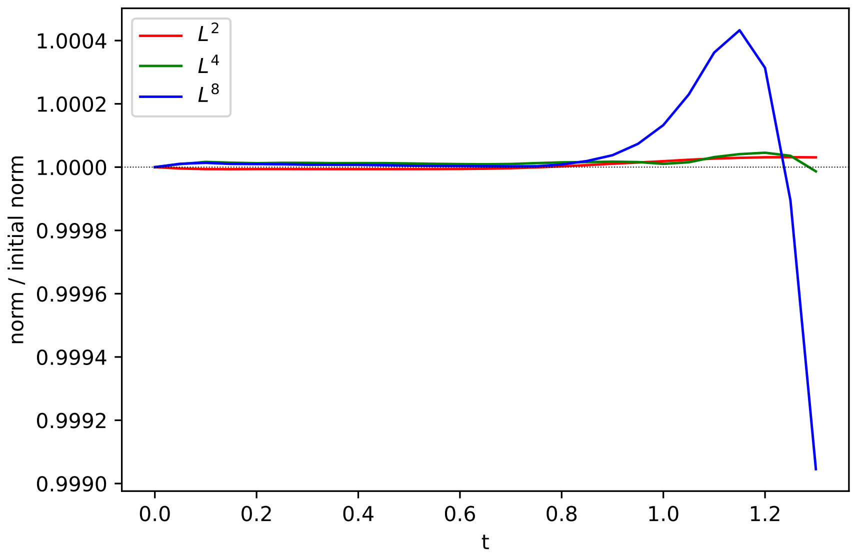

Figure 4: L^p norms normalised to their initial values for $a = 0.5$. All three norms should stay ≈ 1; a sustained drift triggers the resolution-check event that stops the integration.

5.4 Dependence on Scale Parameter a

We run the simulation for eight values of a in the range $[0.5, 0.9375]$ in steps of $1/16$ (matching the range used in the original study). Figure 5 shows the log mix norm for all eight runs. Smaller a (narrower initial support) leads to a faster initial decay but also runs into resolution limitations earlier, since a smaller initial blob is stretched into sub-grid-scale filaments sooner.

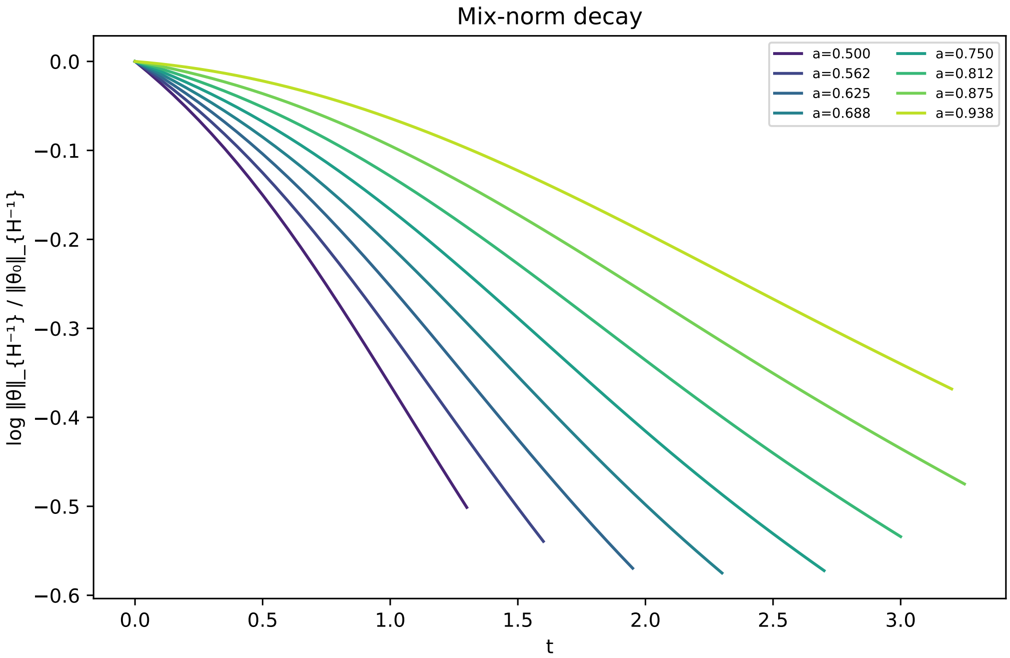

Figure 5: $\log(\|\theta\|_{H^{-1}} \,/\, \|\theta_0\|_{H^{-1}})$ vs. time for $a \in \{0.5, 0.5625, \ldots, 0.9375\}$ (colour coded from dark to light). Each run terminates when the resolution check fires.

5.5 Mixing Timescale vs. Scale Parameter

To quantify the mixing rate for each a, we fit a line to $\log \|\theta\|_{H^{-1}}$ over the *last two-thirds* of each trajectory (discarding the initial transient); the fitted slope is $-r(a)$ with decay-rate magnitude $r(a) > 0$, and the mixing timescale is $\tau(a) = 1/r(a)$.

Table 1 summarises the results. Figure 6 shows $\tau(a)$ plotted against a together with a linear fit (dashed red).

Table 1: Fitted decay-rate magnitudes r and mixing timescales $\tau = 1/r$ for each scale parameter a. ($N = 64$, $F = 1$, initial data: sine-bump family.) These values were regenerated independently for this report and match the values originally recorded for this pipeline to four decimal places in every entry.

a	t_{stop}	$\|\theta(t_{\text{stop}})\|_{H^{-1}}$	r	$\tau = 1/r$
0.5000	1.30	0.6058	0.4400	2.27
0.5625	1.60	0.5831	0.3858	2.59
0.6250	1.95	0.5658	0.3342	2.99
0.6875	2.30	0.5627	0.2863	3.49
0.7500	2.70	0.5642	0.2436	4.11
0.8125	3.00	0.5862	0.2074	4.82
0.8750	3.25	0.6220	0.1741	5.74
0.9375	3.20	0.6920	0.1423	7.03

Two features of Table 1 deserve comment beyond the monotone growth of $\tau(a)$. First, t_{stop} grows with a – larger initial blobs survive longer before the resolution check fires, simply because they take longer to be stretched down to the grid scale. Second, $\|\theta(t_{\text{stop}})\|_{H^{-1}}$ is *not* monotone in a: it decreases from $a = 0.5$ to a minimum around $a \approx 0.6875$–0.75, then increases again toward $a = 0.9375$. This non-monotonicity reflects a competition between two effects that both depend on a: smaller blobs mix faster per unit time (favouring small final mix norm) but are also stopped sooner by the resolution check (favouring a less-mixed final state); the minimum marks where these two effects roughly balance.

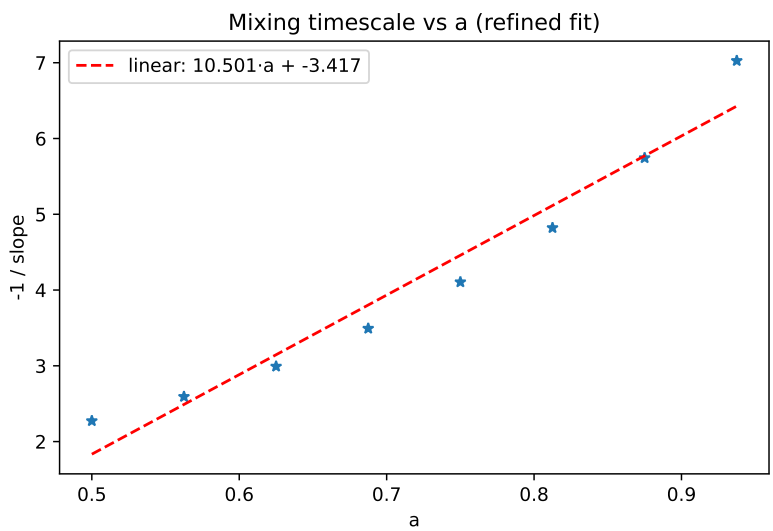

Figure 6: Mixing timescale $\tau(a) = 1/r(a)$ vs. a. The fitted decay-rate magnitude scales as $r \sim a^{-1.78}$ (power-law fit to $\log r$ vs. $\log a$), compared with the a^{-1} scaling characteristic of the Iyer–Kiselev–Xu enstrophy-constrained lower bound [2]. The dashed red line is a linear fit to τ vs. a.

Remark 2. The observed scaling $r \sim a^{-1.78}$ deviates from the a^{-1} dependence characteristic of the rigorous exponential lower bound of Iyer, Kiselev and Xu for enstrophy-constrained mixing of compactly supported two-dimensional data of support area $O(a^2)$ [2]. That result is a lower bound on how fast the H^{-1} norm can possibly decay, not an exact prediction for the dynamically chosen LTD decay rate, so a discrepancy between the two is not automatically evidence of any single mechanism (finite resolution, sub-optimality of the greedy strategy, or both). Longer integration times (larger N to delay the resolution event) may be needed to isolate the asymptotic regime. Section 6 examines this by comparing the fitted exponent at two different resolutions.

6 Numerical Verification and Resolution Sensitivity

This section is not present in Gautam Iyer's earlier MATLAB implementation; it was added to (i) independently verify the reproducibility of Table 1, and (ii) directly test the Remark of Section 5.5 – that the deviation from the a^{-1} scaling may be, in part, a finite-resolution artefact – by rerunning the same eight-value sweep of Section 5.4 at a coarser resolution, $N = 32$, and comparing the fitted results.

6.1 Reproducibility at $N = 64$

Regenerating Table 1 from a clean run of the simulation reproduces every entry to four decimal places: t_{stop}, $\|\theta(t_{\text{stop}})\|_{H^{-1}}$, and r all match exactly, and the fitted power law recovers $r \sim a^{-1.777}$, matching the $a^{-1.78}$ quoted in Section 5.5 to the precision reported there. This confirms deterministic reproducibility of the RK45 integration, the resolution-check event, and the least-squares fitting procedure for the tested software environment and parameter configuration used in this study; it does not establish independence from library versions or execution environments, which were not varied here.

6.2 Resolution Dependence of the Fitted Exponent

Table 2 compares the $N = 64$ results of Table 1 against the same eight-value a-sweep run at $N = 32$, using identical initial data, enstrophy constraint, and fitting procedure.

Table 2: Resolution sensitivity: mixing timescale $\tau(a) = 1/r(a)$ at $N = 32$ versus $N = 64$, same a-sweep and fitting procedure.

a	t_{stop} (N=32)	τ (N=32)	t_{stop} (N=64)	τ (N=64)
0.5000	0.65	2.83	1.30	2.27
0.5625	0.75	3.23	1.60	2.59
0.6250	1.00	3.49	1.95	2.99
0.6875	1.20	3.97	2.30	3.49
0.7500	1.45	4.51	2.70	4.11
0.8125	1.75	5.23	3.00	4.82
0.8750	1.90	6.44	3.25	5.74
0.9375	2.00	8.29	3.20	7.03

Two observations follow directly from Table 2. First, t_{stop} at $N = 32$ is uniformly and substantially smaller than at $N = 64$ – roughly half, across every value of a – confirming that coarser grids hit the resolution limit sooner, exactly as expected since a coarser grid can resolve fewer decades of filamentation before under-resolution contaminates the conserved L^p norms. Second, the fitted power-law exponent itself changes with resolution:

$r \sim a^{-1.61}$ at $N = 32$ versus $r \sim a^{-1.78}$ at $N = 64$. Since both exponents are fitted from trajectories that are truncated early by the same resolution-check mechanism, and the truncation point itself moves with N (Section 3.3), this change from approximately -1.61 to -1.78 demonstrates that the fitted exponent is resolution-dependent and not yet numerically converged. Because the higher-resolution estimate moves *farther* from the benchmark -1 scaling rather than closer to it, these two resolutions alone do not establish that finite-resolution effects explain the discrepancy from the Iyer–Kiselev–Xu benchmark scaling [2]: they show only that at least two resolutions are insufficient to determine the limiting exponent, and cannot distinguish whether higher resolution would approach -1, remain near -1.8, or behave otherwise. Disentangling genuine greedy-strategy behaviour from finite-resolution truncation would require runs at substantially higher N (and correspondingly finer, possibly implicit or exponential, time-stepping to keep the cost tractable), which is identified as future work in Section 7.

7 Conclusions

We have numerically reproduced and extended Gautam Iyer's earlier MATLAB implementation of optimal passive-scalar mixing, adding a first-principles derivation of the LTD optimal velocity (Section 2.5), an explicit account of the numerical scheme's treatment (and non-treatment) of aliasing error (Section 3.3), and a resolution-sensitivity study not present in that earlier implementation (Section 6). The Python pipeline is deterministically reproducible for the tested software environment: a clean rerun reproduces the values reported here to four decimal places in every entry.

The main numerical findings are:

- The H^{-1} mix norm decays approximately exponentially under the LTD optimal velocity, consistent with the theoretical predictions of [1], and consistent with the monotonic-decrease property established rigorously in Proposition 1.

- The mixing timescale $\tau(a) = 1/r(a)$ grows with a, confirming that scalars with wider initial support mix more slowly, while the mix norm reached at the stopping time is non-monotone in a, reflecting a trade-off between faster mixing and earlier resolution loss for smaller a.

- The observed power-law scaling $r \sim a^{-1.78}$ deviates from the a^{-1} dependence characteristic of the Iyer–Kiselev–Xu enstrophy-constrained lower bound; the resolution-sensitivity study of Section 6 shows this exponent is itself resolution-dependent and not yet numerically converged ($a^{-1.61}$ at $N = 32$ vs. $a^{-1.78}$ at $N = 64$, moving *away* from -1 rather than toward it), so these two resolutions cannot determine how much of the deviation, if any, reflects a genuine feature of the greedy strategy versus a finite-resolution artefact.

- The L^p conservation laws provide a robust, physically motivated *a posteriori* diagnostic of resolution loss and a stopping criterion (Section 3.3), but they do not prevent aliasing and are not a substitute for explicit dealiasing: the simulation instead halts automatically once under-resolution becomes detectable, rather than allowing it to accumulate silently.

Future work could explore:

- Higher resolution ($N = 128$ or $N = 256$), ideally combined with explicit 2/3-rule dealiasing, to separate genuine mixing-rate sub-optimality from finite-resolution truncation effects, and to access the true asymptotic regime before the resolution check intervenes.

- Alternative initial data (e.g. random fields, or the diagonal-pair, strip, and trigonometric-polynomial families of Section 4 not swept in Section 5) and their effect on the mixing rate.

- Comparison with non-greedy optimal strategies, e.g. velocities that account for the effect of stirring over a finite time horizon rather than instantaneously, to directly probe the greedy-vs-global optimality question raised in [1].

- A systematic study of the first-order degeneracy identified in Section 2.6, including initial data explicitly engineered to approach $Pg \approx 0$.

References

[1] Z. Lin, J.-L. Thiffeault, and C. R. Doering, Optimal stirring strategies for passive scalar mixing, *Journal of Fluid Mechanics*, 675:465–476, 2011.

[2] G. Iyer, A. Kiselev, and X. Xu, Lower bounds on the mix norm of passive scalars advected by incompressible enstrophy-constrained flows, *Nonlinearity*, 27(5):973–985, 2014.

[3] L. N. Trefethen, *Spectral Methods in MATLAB*, SIAM, 2000.

[4] J. P. Boyd, *Chebyshev and Fourier Spectral Methods*, 2nd edition, Dover Publications, 2001.